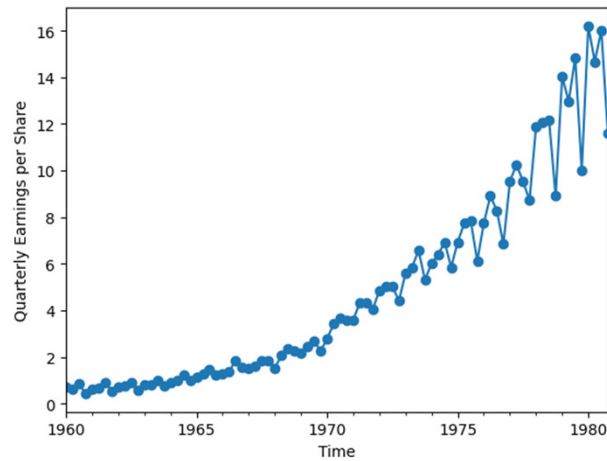


Time Series Model (1)

- Statistics of time series signals (revision)
- AR, MA, ARMA
- Causal, invertibility and redundancy

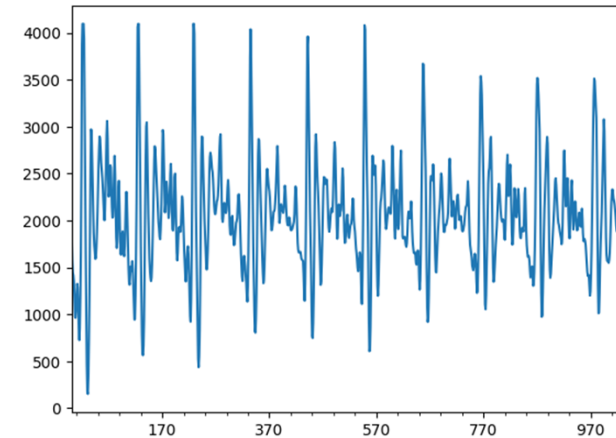
Times series examples.

Annual sale

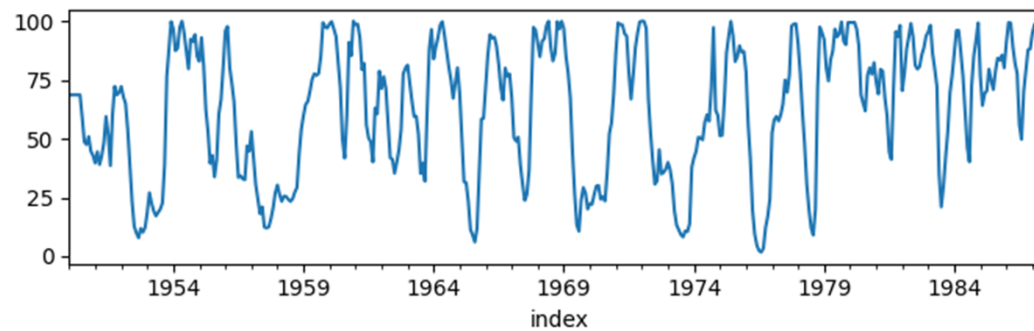


**Maths
models**

Speech



Recruitment



Time series models

Autoregression (AR)

signal at time t

white Gaussian noise

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + w_t$$

AR(p)

Moving average (MA)

$$x_t = w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} + \cdots + \theta_q w_{t-q}$$

MA(q)

ARMA(p, q)

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} + \cdots + \theta_q w_{t-q}$$

Statistics of time series signals

Mean: $\mu_{x_t} = E(x_t) = \int_{-\infty}^{\infty} x f_t(x) dx$

Note: $E(\alpha x_t + \beta y_t + \gamma) = \alpha E(x_t) + \beta E(y_t) + \gamma$

Autocovariance: $\gamma_x(s, t) = \text{cov}(x_s, x_t) = E[(x_s - \mu_{x_s})(x_t - \mu_{x_t})]$

Note: $\text{cov}(\alpha x, \beta y) = \alpha \beta \text{cov}(x, y)$

Variance: $\gamma_x(t, t) = E[(x_t - \mu_{x_t})^2] = \text{var}(x_t)$

Statistics of time series signals (2)

Autocovariance: $\gamma_x(s, t) = \text{cov}(x_s, x_t) = \text{E}[(x_s - \mu_{x_s})(x_t - \mu_{x_t})]$

Autocorrelation function (ACF):

$$\rho(s, t) = \text{corr}(x_s, x_t) = \frac{\gamma_x(s, t)}{\sqrt{\gamma_x(s, s)\gamma_x(t, t)}}$$

$\rho(s, t) = 1$: We can perfectly predict $x_t = \beta_0 + \beta_1 x_s, \beta_1 > 0$.

$= -1$: We can perfectly predict $x_t = \beta_0 + \beta_1 x_s, \beta_1 < 0$.

Statistics of time series signals (3)

Cross-covariance: $\gamma_{xy}(s, t) = \text{cov}(x_s, y_t) = \text{E}[(x_s - \mu_{x_s})(y_t - \mu_{y_t})]$

Cross-correlation function (CCF):

$$\rho_{xy}(s, t) = \text{corr}(x_s, y_t) = \frac{\gamma_{xy}(s, t)}{\sqrt{\gamma_x(s, s)\gamma_y(t, t)}}$$

Strictly Stationary Time Series

“A strictly stationary time series is one for which the probabilistic behavior of every collection of values $\{x_{t_1}, x_{t_2}, \dots, x_{t_k}\}$ is identical to that of the time shifted set $\{x_{t_1+h}, x_{t_2+h}, \dots, x_{t_k+h}\}$. That is

$$\Pr\{x_{t_1} \leq c_1, x_{t_2} \leq c_2, \dots, x_{t_k} \leq c_k\} = \Pr\{x_{t_1+h} \leq c_1, x_{t_2+h} \leq c_2, \dots, x_{t_k+h} \leq c_k\}$$

For all $k = 1, 2, \dots$, all time points $t_1, t_2, \dots, t_k$, all numbers $c_1, c_2, \dots, c_k$, and all time shifts $h = 0, \pm 1, \pm 2, \dots$ ”

Too strict for most application.

(Weakly) Stationary Time Series

“A weakly stationary time series is a finite variance process such that

1. $E(x_t) = E(x_{t+h}) = \mu_x$, for all $h = 0, \pm 1, \pm 2, \dots$
2. $\gamma(s, t)$ is the function of $|s - t|$.”

A stationary process is “weakly stationary,” unless it is stated to be “stationary in the strict sense”,

Autocovariance of stationary process

$$\begin{aligned}\gamma_x(s, t) &= E[(x_s - \mu_{x_s})(x_t - \mu_{x_t})] \\&= E[(x_s - \mu_{x_s})(x_t - \mu_{x_t})] \\&= E[x_s x_t] - \mu_{x_s} E[x_t] - \mu_{x_t} E[x_s] + \mu_{x_s} \mu_{x_t} \\&= E[x_s x_t] - \mu_{x_s} \mu_{x_t} - \mu_{x_t} \mu_{x_s} + \mu_{x_s} \mu_{x_t} \\&= E[x_s x_t] - \mu_{x_s} \mu_{x_t}\end{aligned}$$

Weakly stationary process:

$$\mu_{x_s} = \mu_{x_t} = \mu, \quad E[x_s x_t] = E[x_{s+h} x_{t+h}], \text{ for all } h = 0, \pm 1, \pm 2, \dots$$

$$\gamma_x(s, t) = E[x_{s+h} x_{t+h}] - \mu^2 = \gamma_x(s + h, t + h)$$

Autocovariance of stationary process (2)

$$\gamma_x(s, t) = \gamma_x(s + h, t + h)$$

$$\gamma_x(0, 1) = \gamma_x(0 + h, 1 + h) = \gamma(h, h + 1) = \gamma(1)$$

$$\gamma_x(0, 2) = \gamma_x(0 + h, 2 + h) = \gamma(h, h + 2) = \gamma(2)$$

$\vdots$

$$\gamma_x(0, n) = \gamma_x(0 + h, n + h) = \gamma(h, h + n) = \gamma(n)$$

$$\gamma_x(s, t) = \gamma_x(s - t) = \gamma_x(t - s)$$

White Gaussian Noise (w_t)

- Zero mean

$$E(w_t) = 0, \text{ for all } t.$$

- Independent with variance σ_w^2

$$\gamma_w(0) = E(w_t^2) = \sigma_w^2$$

$$\gamma_w(h) = E(w_t w_{t+h}) = E(w_t)E(w_{t+h}) = 0, \text{ for all } h \neq 0.$$

$$\gamma_w(h) = \begin{cases} \sigma_w^2, & h = 0 \\ 0, & \text{otherwise} \end{cases}$$

w_t is stationary.

Ex. 1: Statistics of AR(1) process

Let x_t be a stationary process and $w_t = N(\mathbf{0}, \sigma_w^2)$.

$$x_t = \phi x_{t-1} + w_t, \quad |\phi| < 1$$

Solution

Mean: $E(x_t) = E(\phi x_{t-1} + w_t) = \phi E(x_{t-1}) + \mathbf{0}$

$$\mu_x = \phi \mu_x \therefore \mu_x = 0$$

Covariance: $\gamma(n) = E((x_t - \mu_x)(x_{t+n} - \mu_x)) = E(x_t x_{t+n})$

Ex. 1: Variance of AR(1) process

$$x_t = \phi x_{t-1} + w_t$$

Solution

$$\begin{aligned}\gamma(0) &= E(x_t x_t) = E(x_t^2) = E((\phi x_{t-1} + w_t)^2) \\ &= E(\phi^2 x_{t-1}^2 + 2\phi x_{t-1} w_t + w_t^2) \\ &= \phi^2 E(x_{t-1}^2) + 2\phi E(x_{t-1}) E(w_t) + E(w_t^2) \\ &= \phi^2 \gamma(0) + 0 + \sigma_w^2\end{aligned}$$

$$\gamma(0) = \frac{\sigma_w^2}{1 - \phi^2}$$

Ex. 1: Covariance of AR(1) process

$$x_t = \phi x_{t-1} + w_t$$

Solution: Let $h > 0$.

$$\begin{aligned}\gamma(h) &= E(x_t x_{t+h}) \\ &= E(x_t (\phi x_{t+h-1} + w_{t+h})) \quad = 0 \text{ for } h > 0 \\ &= \phi E(x_t x_{t+h-1}) + \boxed{E(w_{t+h} x_t)} \\ &= \phi \gamma(h-1) \text{ for } h > 0\end{aligned}$$

$$\gamma(1) = \phi \gamma(0)$$

$$\gamma(2) = \phi \gamma(1) = \phi(\phi \gamma(0)) = \phi^2 \gamma(0)$$

$$\gamma(3) = \phi \gamma(2) = \phi(\phi^2 \gamma(0)) = \phi^3 \gamma(0)$$

$\vdots$

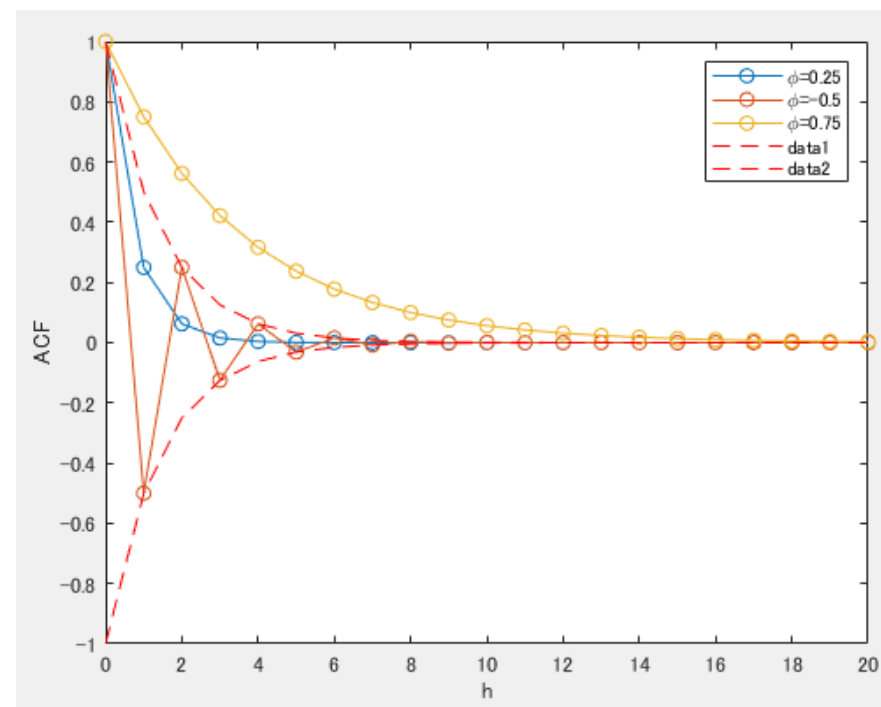
Times Series Models

$$\left. \begin{array}{l} \gamma(1) = \phi \gamma(0) \\ \gamma(2) = \phi \gamma(1) = \phi(\phi \gamma(0)) = \phi^2 \gamma(0) \\ \gamma(3) = \phi \gamma(2) = \phi(\phi^2 \gamma(0)) = \phi^3 \gamma(0) \\ \vdots \end{array} \right\} \gamma(h) = \phi^h \gamma(0)$$

Ex. 1: Covariance and ACF of AR(1)

$$\gamma(h) = \frac{\phi^h \sigma_w^2}{1 - \phi^2}$$

$$\text{ACF} = \rho(h) = \frac{\gamma(h)}{\gamma(0)} = \phi^h$$



Ex. 2: Statistics of MA(1) process

Let w_t be the zero-mean white Gaussian noise and $N(\mathbf{0}, \sigma_w^2)$.

$$x_t = w_t + \theta w_{t-1}$$

Solution

$$\text{Mean: } \mu_x = E(x_t) = E(w_t + \theta w_{t-1}) = E(w_t) + \theta E(w_{t-1}) = 0$$

$$\text{Covariance: } \gamma(h) = E((x_t - \mu_x)(x_{t+h} - \mu_x)) = E(x_t x_{t+h})$$

$$\gamma_w(h) = \begin{cases} \sigma_w^2, & h = 0 \\ 0, & \text{otherwise} \end{cases}$$

Ex. 2: Variance of MA(1) process.

$$x_t = w_t + \theta w_{t-1}$$

Solution

$$\begin{aligned} \gamma(0) &= E(x_t x_t) = E(x_t^2) = E((w_t + \theta w_{t-1})^2) \\ &= E(w_t^2 + 2\theta w_t w_{t-1} + \theta^2 w_{t-1}^2) \\ &= E(w_t^2) + 2\theta E(w_t w_{t-1}) + \theta^2 E(w_{t-1}^2) \\ &= \gamma_w(0) + 2\theta \gamma_w(1) + \theta^2 \gamma_w(0) \\ &= \sigma_w^2 + 2\theta(0) + \theta^2 \sigma_w^2 \\ &= (1 + \theta^2) \sigma_w^2 \end{aligned}$$

$$\gamma_w(h) = \begin{cases} \sigma_w^2, & h = 0 \\ 0, & \text{otherwise} \end{cases}$$

Ex. 2: Covariance of MA(1) process

$$x_t = w_t + \theta w_{t-1}$$

Solution

$$\begin{aligned} \gamma(1) &= E(x_t x_{t+1}) = E((w_t + \theta w_{t-1})(w_{t+1} + \theta w_t)) \\ &= E(w_t w_{t+1} + \theta w_{t-1} w_{t+1} + \theta w_t^2 + \theta^2 w_{t-1} w_t) \\ &= E(w_t w_{t+1}) + \theta E(w_{t-1} w_{t+1}) + \theta E(w_t^2) + \theta^2 E(w_{t-1} w_t) \\ &= \gamma(1) + \theta \gamma(2) + \theta \gamma(0) + \theta^2 \gamma(1) \\ &= 0 + \theta(0) + \theta(\sigma_w^2) + \theta^2(0) = \theta \sigma_w^2 \end{aligned}$$

$$\gamma_w(h) = \begin{cases} \sigma_w^2, & h = 0 \\ 0, & \text{otherwise} \end{cases}$$

Ex. 2: Covariance of MA(1) process (2)

$$x_t = w_t + \theta w_{t-1}$$

Solution: for $h = 2, 3, 4, \dots$

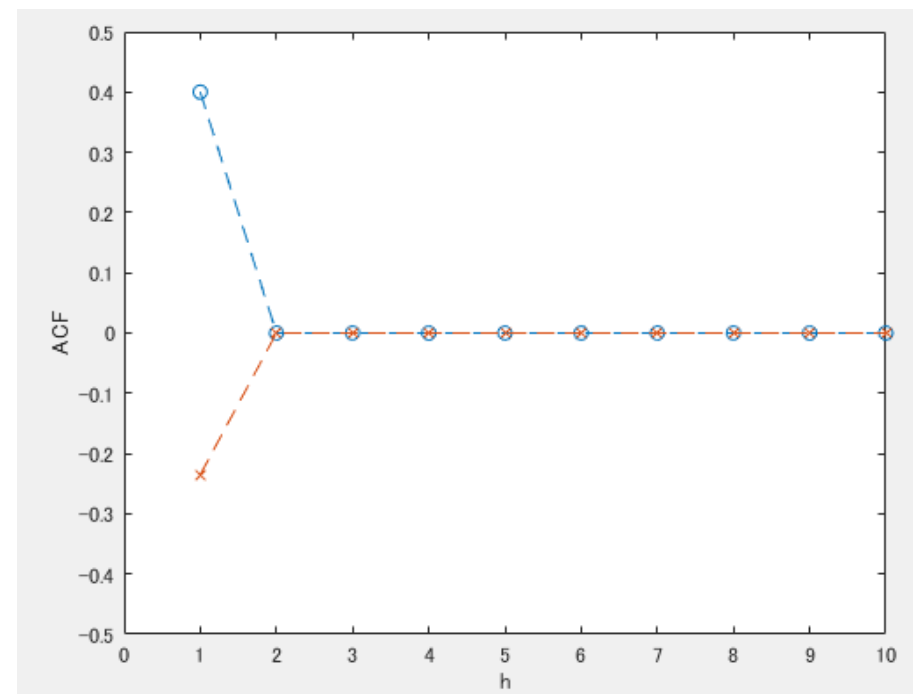
$$\begin{aligned} \gamma(h) &= E(x_t x_{t+h}) = E((w_t + \theta w_{t-1})(w_{t+h} + \theta w_{t+h-1})) \\ &= E(w_t w_{t+h} + \theta w_{t-1} w_{t+h} + \theta w_t w_{t+h-1} + \theta^2 w_{t-1} w_{t+h-1}) \\ &= E(w_t w_{t+h}) + \theta E(w_{t-1} w_{t+h}) + \theta E(w_t w_{t+h-1}) + \theta^2 E(w_{t-1} w_{t+h-1}) \\ &= \gamma_w(h) + \theta \gamma_w(h+1) + \theta \gamma_w(h-1) + \theta^2 \gamma_w(h) \\ &= 0 \end{aligned}$$

Ex. 2: Covariance and ACF of MA(1)

$$\gamma(h) = \begin{cases} (1 + \theta^2)\sigma_w^2, & h = 0 \\ \theta\sigma_w^2, & h = 1 \\ 0, & h > 1 \end{cases}$$

$$\text{ACF} = \rho(h) = \frac{\gamma(h)}{\gamma(0)} = \begin{cases} 1, & h = 0 \\ \frac{\theta}{1+\theta^2}, & h = 1 \\ 0, & h > 1 \end{cases}$$

Note: $\gamma(-h) = \gamma(h)$



Ex. 3: $\gamma(h)$ of MA(q) process

$$x_t = w_t + \sum_{k=1}^q \theta_k w_{t-k} = \sum_{k=0}^q \theta_k w_{t-k}, \theta_0 = 1$$

Solution:

$$\begin{aligned}\gamma(h) &= E[x_t x_{t+h}] = E\left[\left(\sum_{k=0}^q \theta_k w_{t-k}\right)\left(\sum_{l=0}^q \theta_l w_{t+h-l}\right)\right] \\ &= E\left[\sum_{k=0}^q \sum_{l=0}^q \theta_k \theta_l w_{t-k} w_{t+h-l}\right] \\ &= \sum_{k=0}^q \sum_{l=0}^q \theta_k \theta_l E[w_{t-k} w_{t+h-l}] \\ &= \sum_{k=0}^q \sum_{l=0}^q \theta_k \theta_l \gamma_w(l - h - k)\end{aligned}$$

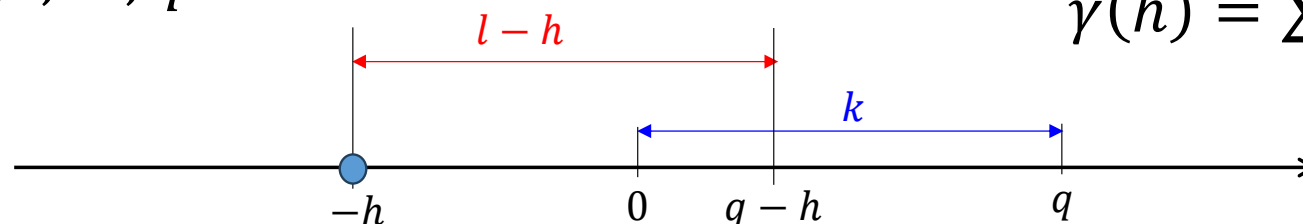
Ex. 3: $\gamma(h)$ of MA(q) process

$$\gamma_w(h) = \begin{cases} \sigma_w^2, & h = 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\gamma(h) = \sum_{k=0}^q \sum_{l=0}^q \theta_k \theta_l \gamma_w(l - h - k)$$

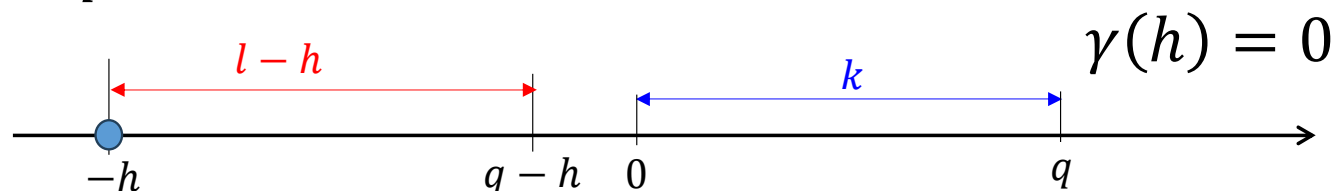
$$\gamma_w(l - h - k) \neq 0 \text{ when } k = l - h$$

for $h = 0, 1, 2, \dots, q$



$$\gamma(h) = \sum_{k=0}^{q-h} \theta_k \theta_{k+h} \sigma_w^2$$

for $h = q + 1, q + 2, \dots$



$$\gamma(h) = 0$$

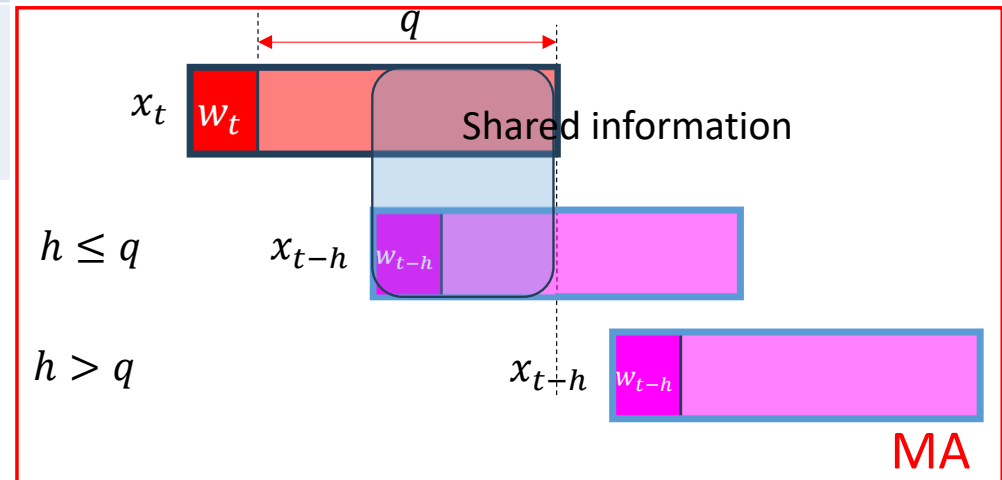
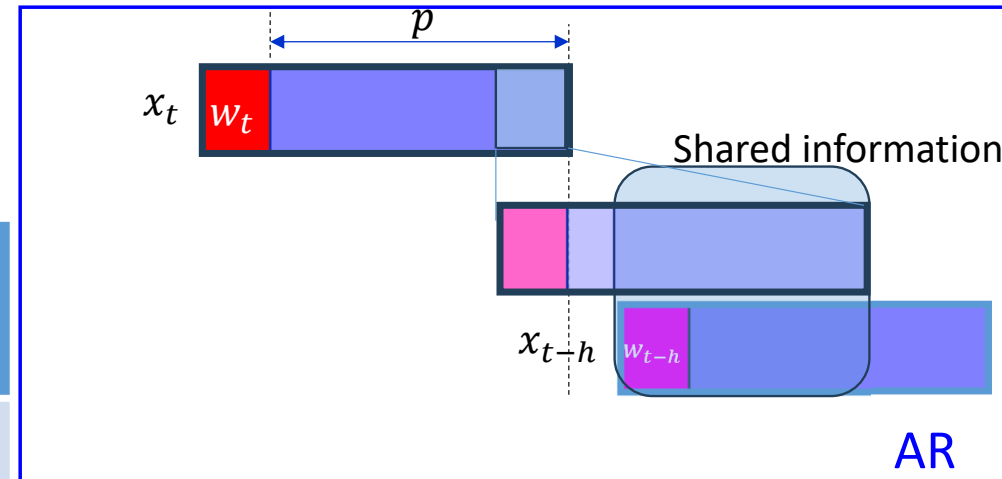
Ex. 3: $\gamma(h)$ of MA(q) process

$$\gamma(h) = \begin{cases} \sum_{k=0}^{q-h} \theta_k \theta_{k+h} \sigma_w^2, & h = 0, 1, 2, \dots, q \\ 0, & h > q \end{cases}$$

$\gamma(h)$ of MA(q) processes is zero when $|h| > |q|$.

Partial summary

Process	$ACF(\rho(h))$
$AR(p)$	Tails off
$MA(q)$	0 when $h > q$



Conversion of AR model to MA model

$$x_t = \sum_{k=1}^p \phi_k x_{t-k} + w_t \quad \longrightarrow \quad x_t = w_t + \sum_{k=1}^{\infty} \psi_k w_{t-k}$$

Idea: AR(1)

$$x_t = \phi x_{t-1} + w_t = \phi(\phi x_{t-2} + w_{t-1}) + w_t = \phi^2 x_{t-2} + w_t + \phi w_{t-1}$$

$x_{t-1} = \phi x_{t-2} + w_{t-1}$

$$= \phi^2(\phi x_{t-3} + w_{t-2}) + w_t + \phi w_{t-1} = \phi^3 x_{t-3} + w_t + \phi w_{t-1} + \phi^2 w_{t-2}$$

$$\vdots$$

$$= \phi^k x_{t-k} + \sum_{j=0}^{k-1} \phi^j w_{t-j} \quad \dots \quad = \sum_{j=0}^{\infty} \phi^j w_{t-j}$$

Tim

How about AR(2)?

Backshift operator (B)

$$Bx_t = x_{t-1}$$

$$B^2x_t = B[Bx_t] = B[x_{t-1}] = x_{t-2}$$

$$B^3x_t = B[B^2x_t] = B[x_{t-2}] = x_{t-3}$$

$$\vdots$$

$$B^nx_t = x_{t-n}$$

Note: Opposite to E operator in 204 Diff Eqn&Transform.

AR and MA models with B operator

AR

$$\begin{aligned} x_t &= \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + w_t \\ &= \phi_1 B x_t + \phi_2 B^2 x_t + \cdots + \phi_p B^p x_t + w_t \\ &= (\phi_1 B + \phi_2 B^2 + \cdots + \phi_p B^p) x_t + w_t \end{aligned}$$

$$(1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p) x_t = w_t$$

$$\phi(B) x_t = w_t$$

MA

$$\begin{aligned} x_t &= w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} + \cdots + \theta_q w_{t-q} \\ &= w_t + \theta_1 B w_t + \theta_2 B^2 w_t + \cdots + \theta_q B^q w_t \\ &= (1 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q) w_t \end{aligned}$$

$$x_t = \theta(B) w_t$$

$$x_t = w_t + \sum_{k=1}^{\infty} \psi_k w_{t-k} = \psi(B) w_t$$

$$w_t = x_t + \sum_{k=1}^{\infty} \pi_k x_{t-k} = \pi(B) x_t$$

Conversion by matching coefficient

$$\phi(B)\boxed{x_t} = w_t$$

$x_t = \psi(B)w_t$

$$\phi(B)\psi(B)w_t = w_t$$

$$(1 - \phi_1 B - \phi_2 B^2 - \dots)(1 + \psi_1 B + \psi_2 B^2 + \dots)w_t = w_t$$

w_{t-1} term w_{t-2} term

$$(1 + \boxed{(\psi_1 - \phi_1)B} + \boxed{(\psi_2 - \psi_1\phi_1 - \phi_2)B^2} - \dots)w_t = w_t$$

$$\psi_1 = \phi_1$$

$$\psi_2 = \psi_1\phi_1 + \phi_2$$

$\vdots$

Ex. 4: MA(1) to AR model conversion

$$\begin{aligned}x_t &= w_t + \theta_1 w_{t-1} \\&= \theta(B) \boxed{w_t} \quad \boxed{w_t = \pi(B)x_t} \\&= \theta(B) \pi(B) x_t \\&= (1 + \theta_1 B) (1 + \pi_1 B + \pi_2 B^2 + \cdots) x_t \\&= (1 + (\pi_1 + \theta_1) B + (\pi_2 + \theta_1 \pi_1) B^2 + (\pi_3 + \theta_1 \pi_2) B^3 + \cdots) x_t \\&\quad \therefore \pi_n = -\theta_1 \pi_{n-1}, \quad \pi_0 = 1 \\&\quad w_t = (1 - \theta_1 B + \theta_1^2 B^2 - \theta_1^3 B^3 + \cdots) x_t \\&\quad x_t = w_t + \theta_1 x_{t-1} - \theta_1^2 x_{t-2} + \theta_1^3 x_{t-3} - \cdots\end{aligned}$$

$$\sum_{k=0}^{\infty} r^k = 1 + r + r^2 + \dots = \frac{1}{1-r} = (1-r)^{-1}, \quad |r| < 1$$

Conversion by inverse operation

$$\begin{aligned} x_t &= w_t + \theta_1 w_{t-1} \\ &= (1 + \theta_1 B) w_t \end{aligned}$$

$$(1 + \theta_1 B)^{-1} x_t = (1 + \theta_1 B)^{-1} (1 + \theta_1 B) w_t$$

$$(1 - \theta_1 B + \theta_1^2 B^2 - \theta_1^3 B^3 + \dots) x_t = w_t$$

$$x_t = w_t + \theta_1 x_{t-1} - \theta_1^2 x_{t-2} + \theta_1^3 x_{t-3} - \dots$$

x_t and y_t are considered the same if they have the same $\gamma(h)$.

Ex. 5: Non-uniqueness of MA models

Find the MA(1) model having $\gamma(h) = \begin{cases} 5, & h = 0 \\ 2, & h = 1. \\ 0, & h > 1 \end{cases}$

Solution: From Ex.2, we have $\gamma(h) = \begin{cases} (1 + \theta^2)\sigma_w^2, & h = 0 \\ \theta\sigma_w^2, & h = 1 \\ 0, & h > 1 \end{cases}$

$$\rho(1) = \frac{\gamma(1)}{\gamma(0)} = \frac{\theta}{1 + \theta^2} = \frac{2}{5} \Rightarrow 2\theta^2 - 5\theta + 2 = 0 \Rightarrow \theta = 0.5, 2$$

When $\theta = 0.5$, $0.5\sigma_w^2 = 2$. We have $x_t = w_t + 0.5w_{t-1}$, $w_t \sim iid N(0,4)$.

When $\theta = 2$, $2\sigma_w^2 = 2$. We have $y_t = w_t + 2w_{t-1}$, $w_t \sim iid N(0,1)$.

Ex. 5: Invertible MA model

Solution: Convert to AR model and choose the one that has infinite representation.

From Ex. 4, we have $x_t = w_t + \theta w_{t-1} = w_t - \sum_{k=1}^{\infty} (-\theta)^k x_{t-k}$

$$x_t = w_t + 0.5w_{t-1} = w_t - \sum_{k=1}^{\infty} \boxed{(-0.5)^k} x_{t-k} \quad \checkmark$$

$\lim_{k \rightarrow \infty} (-0.5)^k = 0.$

$$y_t = w_t + 2w_{t-1} = w_t - \sum_{k=1}^{\infty} \boxed{(-2)^k} y_{t-k} \quad \times$$

$\lim_{k \rightarrow \infty} (-2)^k \rightarrow \infty.$

ARMA model: parameter redundancy

$$\begin{aligned}x_t &= \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} + \cdots + \theta_q w_{t-q} \\&= \phi_1 B x_t + \phi_2 B^2 x_t + \cdots + \phi_p B^p x_t + w_t + \theta_1 B w_t + \theta_2 B^2 w_t + \cdots + \theta_q B^q w_t \\(1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p) x_t &= (1 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q) w_t \\ \phi(B) x_t &= \theta(B) w_t\end{aligned}$$

Multiply $\eta(B)$ to both sides.

$$\eta(B) \phi(B) x_t = \eta(B) \theta(B) w_t$$

New ARMA model with higher p and q .

Ex. 6: Parameter redundancy

Simplify the following process.

$$x_t = \frac{1}{4}x_{t-1} + w_t - \frac{3}{4}w_{t-1} + \frac{1}{8}w_{t-2}$$

Solution: Convert to $\phi(B)x_t = \theta(B)w_t$. Then remove the common factor in $\phi(B)$ and $\theta(B)$.

$$\left(1 - \frac{1}{4}B\right)x_t = \left(1 - \frac{3}{4}B + \frac{1}{8}B^2\right)w_t = \left(1 - \frac{1}{4}B\right)\left(1 - \frac{1}{2}B\right)w_t$$

$$x_t = \left(1 - \frac{1}{2}B\right)w_t = w_t - \frac{1}{2}w_t$$

AR and MA polynomials.

ARMA Model: $\phi(B)x_t = \theta(B)w_t$

“The AR and MA polynomials are defined as

$$\phi(z) = 1 - \phi_1 z - \phi_2 z^2 - \dots - \phi_p z^p, \quad \phi_p \neq 0$$

and

$$\theta(z) = 1 - \theta_1 z - \theta_2 z^2 - \dots - \theta_q z^q, \quad \theta_q \neq 0,$$

respectively, where z is a complex number.”

***** z becomes the delay operator instead of z^{-1} .*****

ARMA(p, q) model: definition

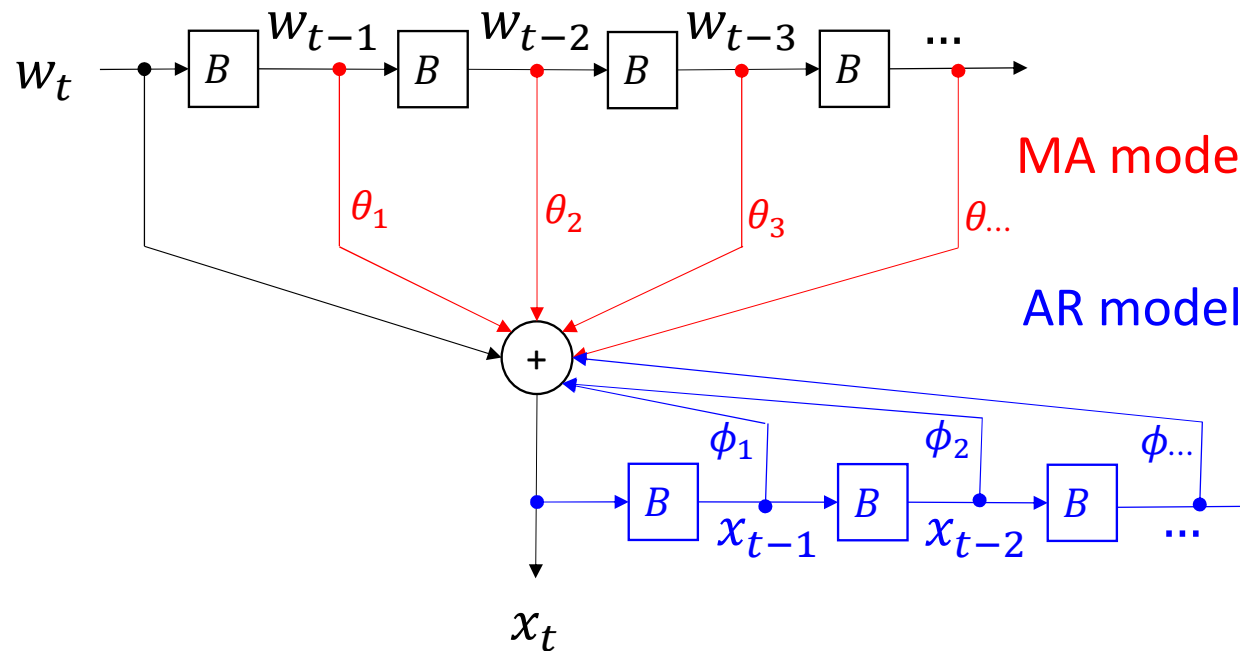
“A time series $\{x_t; t = 0, \pm 1, \pm 2, \dots\}$ is ARMA(p, q) if it is stationary and

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \dots + \phi_p x_{t-p} + w_t + \theta_1 w_{t-1} + \theta_2 w_{t-2} + \dots + \theta_q w_{t-q},$$

with $\phi(z)$ and $\theta(z)$ has no common factors, $\phi_p \neq 0, \theta_q \neq 0$ and $\sigma_w^2 > 0$. The parameter p and q are the autoregressive and the moving average orders, respectively.”

Note: If $E(x_t) = \mu$, replace x_{t-k} with $x_{t-k} - \mu$ where $k = 0, \pm 1, \pm 2, \dots$

Causality (stability) of ARMA model



MA model = feedforward system
 → always stable

AR model = feedback loop
 → require stability

$$x_t = w_t + \sum_{k=1}^q \theta_k w_{t-k} + \sum_{l=1}^p \phi_l x_{t-l}$$

Causality of ARMA model (2)

$$x_t = w_t + \sum_{k=1}^q \theta_k w_{t-k} + \sum_{l=1}^p \phi_l x_{t-l} = \sum_{j=0}^{\infty} \psi_j w_{t-j}$$

This infinite series must converge.

“An **ARMA(p,q) model** is said to be **causal**, if the time series $\{x_t; t = 0, \pm 1, \pm 2, \dots\}$ can be written as one-sided linear process:

$$x_t = \sum_{j=0}^{\infty} \psi_j w_{t-j} = \psi(B)w_t,$$

where $\psi(B) = \sum_{j=0}^{\infty} \psi_j B^j$ and $\sum_{j=0}^{\infty} |\psi_j| < \infty$ where $\psi_0 = 1$.

Invertibility of ARMA model

$$x_t = w_t + \sum_{k=1}^q \theta_k w_{t-k} + \sum_{l=1}^p \phi_l x_{t-l} = w_t + \sum_{j=1}^{\infty} \pi_j x_{t-j}$$

This infinite series must converge.

“An **ARMA(p,q) model** is said to be **invertible**, if the time series $\{x_t; t = 0, \pm 1, \pm 2, \dots\}$ can be written as

$$\pi(B)x_t = \sum_{j=0}^{\infty} \pi_j x_{t-j} = w_t,$$

where $\pi(B) = \sum_{j=0}^{\infty} \pi_j B^j$ and $\sum_{j=0}^{\infty} |\pi_j| < \infty$ where $\pi_0 = 1$.

Poles and zeros in ARMA model

$$\phi(B)x_t = \theta(B)w_t$$

$$x_t = \frac{\theta(B)}{\phi(B)}w_t = \psi(B)w_t$$

Rewrite $\psi(B)$ in term of AR and MA polynomial. $\psi(z) = \frac{\theta(z)}{\phi(z)}$

1. Causal (stable) if the poles ($\phi(z) = 0$) lies **outside** the unit circle.
2. Invertible if the zeros ($\theta(z) = 0$) lies **outside** the unit circle.

Ex. 7: Non-causal AR(1) process

Rewrite the following non-causal AR(1) model to the stable infinite series.

$$x_t = 2x_{t-1} + w_t$$

Solution: Formulate x_t as the function of x_{t+1} and w_{t+1} .

$$x_{t+1} = 2x_t + w_{t+1} \quad \therefore x_t = 0.5x_{t+1} - 0.5w_{t+1}$$

$$x_t = 0.5 \boxed{x_{t+1}} - 0.5w_{t+1} = 0.5 \boxed{(0.5x_{t+2} - 0.5w_{t+2})} - 0.5w_{t+1}$$

$$\boxed{x_{t+1} = 0.5x_{t+2} - 0.5w_{t+2}}$$

$$= 0.5^2 x_{t+2} - 0.5^2 w_{t+2} - 0.5w_{t+1}$$

$$= 0.5^2 (0.5x_{t+3} - 0.5w_{t+3}) - 0.5^2 w_{t+2} - 0.5w_{t+1}$$

$$= 0.5^3 x_{t+3} - (0.5^3 w_{t+3} + 0.5^2 w_{t+2} + 0.5w_{t+1})$$

Ex.7: Non-causal AR(1) process

$$\begin{aligned} x_t &= 0.5^3 x_{t+3} - (0.5^3 w_{t+3} + 0.5^2 w_{t+2} + 0.5 w_{t+1}) \\ &\vdots \\ x_t &= 0.5^k x_{t+k} - \sum_{j=1}^k 0.5^j w_{t+j} = - \sum_{j=1}^{\infty} 0.5^j w_{t+j} \end{aligned}$$

Non-causal model depends on the future!!!

Ex. 7: Conversion to causal AR(1) process

$$x_t = 2x_{t-1} + w_t$$

Solution: Find AR(1) process that have the same $\gamma(h)$ as x_t .

****Write $x(t)$ as the MA process depended on the future w_t .**

Let $x_t = \sum_{k=0}^{\infty} \psi_k w_{t+k}$.

$$\sum_{k=0}^{\infty} \psi_k w_{t+k} = 2 \sum_{k=0}^{\infty} \psi_k w_{t-1+k} + w_t$$

$$-2\psi_0 w_{t-1} + \sum_{k=0}^{\infty} (\psi_k - 2\psi_{k+1}) w_{t+k} = w_t$$

Ex. 7: Conversion to causal AR(1) process (2)

$$-2\psi_0 w_{t-1} + \sum_{k=0}^{\infty} (\psi_k - 2\psi_{k+1}) w_{t+k} = w_t$$

$$\psi_0 = 0$$

$$\psi_0 - 2\psi_1 = 1 \implies \psi_1 = -0.5$$

$$\psi_{k+1} = 0.5\psi_k \implies \psi_2 = 0.5\psi_1 = -(0.5^2)$$

$$\psi_3 = 0.5\psi_2 = -(0.5^3)$$

$$\vdots$$

$$\psi_k = -(0.5^k)$$

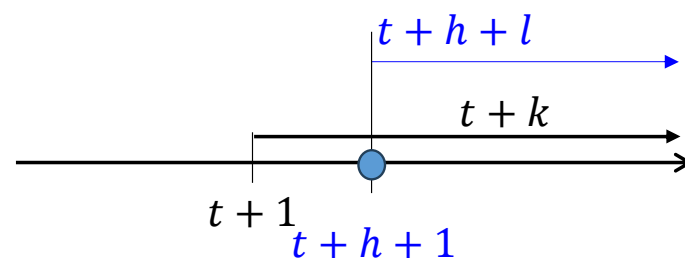
$$x_t = - \sum_{k=1}^{\infty} 0.5^k w_{t+k}$$

Ex. 7: Conversion to causal AR(1) process (3)

$$x_t = - \sum_{k=1}^{\infty} 0.5^k w_{t+k}$$

$$\gamma(h) = E((x_t - \mu_x)(x_{t+h} - \mu_x)) = E(x_t x_{t+h})$$

$$= E \left[- \sum_{k=1}^{\infty} 0.5^k w_{t+k} \left[- \sum_{l=1}^{\infty} 0.5^l w_{t+h+l} \right] \right]$$



$$= \sum_{k=1}^{\infty} \sum_{l=1}^{\infty} 0.5^{k+l} E[w_{t+k} w_{t+h+l}] = \sum_{l=1}^{\infty} 0.5^{2l+h} \sigma_w^2 = 0.5^h \sigma_w^2 \sum_{l=1}^{\infty} [0.5^2]^l$$

Ex. 7: Conversion to causal AR(1) process (4)

$$\gamma(h) = 0.5^h \sigma_w^2 \sum_{l=1}^{\infty} [0.5^2]^l = 0.5^h \sigma_w^2 \left(\frac{0.5^2}{1 - 0.5^2} \right) = \frac{0.5^h}{1 - 0.5^2} [0.5^2 \sigma_w^2]$$

From Ex. 2, $\gamma(h)$ of AR(1) process is $\gamma(h) = \frac{\phi^h \sigma_w^2}{1 - \phi^2}$.

By matching coefficient, we have $\phi = 0.5$ and σ_w^2 of the causal process is the quarter of the non-causal model.

The causal model is $x_t = 0.5x_{t-1} + w_{t,new}$, $w_{t,new} \sim iid N(0, 0.25\sigma_w^2)$.

Ex.8: Parameter redundancy, causality and invertibility

Simplify the following process to causal and invertible process.

$$x_t = 0.4x_{t-1} + 0.45x_{t-2} + w_t + w_{t-1} + 0.25w_{t-2}$$

Solution: Simplify by removing the common factor in $\phi(B)x_t = \theta(B)w_t$.

$$(1 - 0.4B - 0.45B^2)x_t = (1 + B + 0.25B^2)w_t$$

$$(1 - 0.9B)(1 + 0.5B)x_t = (1 + 0.5B)^2w_t$$

$$(1 - 0.9B)x_t = (1 + 0.5B)w_t$$

Ex.8: Causality and invertibility

$$(1 - 0.9B)x_t = (1 + 0.5B)w_t$$

$$x_t = \frac{(1 + 0.5B)}{(1 - 0.9B)} w_t$$

Causality: the poles of $\psi(z)$ is outside the unit circle.

$$(1 - 0.9z) = 0 \quad \therefore |z| = \left| \frac{1}{0.9} \right| > 1 \quad \checkmark$$

Invertibility: the zeros of $\psi(z)$ is outside the unit circle.

$$(1 + 0.5z) = 0 \quad \therefore |z| = \left| \frac{-1}{0.5} \right| > 1 \quad \checkmark$$

The causal and invertible simplified process is $x_t = 0.9x_{t-1} + w_t + 0.5w_{t-1}$.

Ex. 9: Causality condition of AR(2) process

Find the causal condition of the AR(2) process.

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + w_t$$

Solution: The poles of $\psi(z)$, where $x_t = \psi(B)w_t$, must be outside the unit circle.

$$(1 - \phi_1 B - \phi_2 B^2)x_t = w_t$$
$$x_t = \frac{1}{(1 - \phi_1 B - \phi_2 B^2)} w_t$$

$$(1 - \phi_1 z - \phi_2 z^2) = 0$$

$$z^{-2} - \phi_1 z^{-1} - \phi_2 = 0$$

Ex. 9: Causality condition of AR(2) process (2)

$$\lambda^2 - \phi_1 \lambda - \phi_2 = 0 \quad \therefore \lambda = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2}$$

$$z_1^{-1} = \lambda_1 = \frac{\phi_1 + \sqrt{\phi_1^2 + 4\phi_2}}{2}, \quad z_2^{-1} = \lambda_2 = \frac{\phi_1 - \sqrt{\phi_1^2 + 4\phi_2}}{2}$$

For the real roots, $\phi_1^2 + 4\phi_2 \geq 0$.

$$\left| \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2} \right| < 1 \quad \Rightarrow \quad \left| \phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2} \right| < 2$$

Ex. 9: Causality condition of AR(2) process (3)

$$\left| \phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2} \right| < 2$$

When $\phi_1 > 0$, $|z_1|$ is larger and the term inside the absolute is positive.

$$\phi_1 + \sqrt{\phi_1^2 + 4\phi_2} < 2$$

$$\sqrt{\phi_1^2 + 4\phi_2} > 2 - \phi_1$$

$$\phi_1^2 + 4\phi_2 > 4 - 4\phi_1 + \phi_1^2$$

$$\phi_2 + \phi_1 > 1$$

Ex. 9: Causality condition of AR(2) process (4)

$$\left| \phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2} \right| < 2$$

When $\phi_1 < 0$, $|z_2|$ is larger and the term inside the absolute is negative. $|z_2|$ is smaller than 2 so $-2 < z_2$. In the similar manner, we have

$$\phi_2 - \phi_1 < 1.$$

Because $|z_1|, |z_2| > 1$, we have $|\lambda_1|, |\lambda_2| < 1$ so

$$|\lambda_1||\lambda_2| = \left| \frac{\phi_1 + \sqrt{\phi_1^2 + 4\phi_2}}{2} \right| \left| \frac{\phi_1 - \sqrt{\phi_1^2 + 4\phi_2}}{2} \right| < 1$$

$$|\phi_1^2 - (\phi_1^2 + 4\phi_2)| < 4 \quad \Rightarrow \quad |\phi_2| < 1$$

Ex. 9: Causality condition of AR(2) process (5)

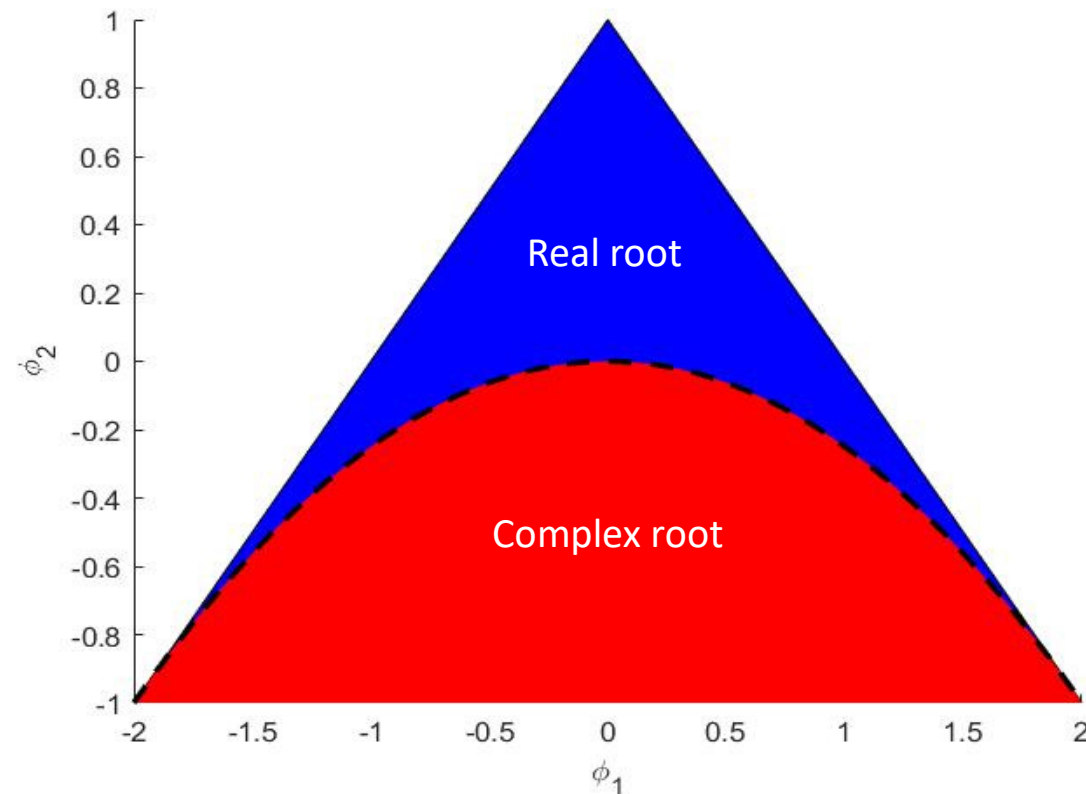
For the complex roots, $\phi_1^2 + 4\phi_2 < 0$. It is clear that $\phi_2 < 0$.

$$z^{-1} = \lambda = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2} = \frac{\phi_1 \pm j\sqrt{-(\phi_1^2 + 4\phi_2)}}{2}$$

$$1 > |z^{-1}|^2 = \frac{\phi_1^2 + \left(-(\phi_1^2 + 4\phi_2)\right)}{4} = -\phi_2$$

$$-1 < \phi_2 < 0$$

Ex. 9: Causality condition of AR(2) process (5)



Autocovariance of causal ARMA(p, q) process

$$\begin{aligned}x_t &= \sum_{k=1}^p \phi_k x_{t-k} + w_t + \sum_{l=1}^q \theta_l w_{t-l} \\&= \sum_{k=1}^p \phi_k x_{t-k} + \sum_{l=0}^q \theta_l w_{t-l}, \quad \theta_0 = 1\end{aligned}$$

$$\begin{aligned}\gamma(h) &= E(x_t x_{t+h}) = E\left(x_t \left[\sum_{k=1}^p \phi_k x_{t+h-k} + \sum_{l=0}^q \theta_l w_{t+h-l} \right]\right) \\&= \sum_{k=1}^p \phi_k \gamma(h-k) + \sum_{l=0}^q \theta_l E(x_t w_{t+h-l})\end{aligned}$$

Autocovariance of causal ARMA(p, q) process (2)

$$\begin{aligned}
 \gamma(h) &= \sum_{k=1}^p \phi_k \gamma(h-k) + \sum_{l=0}^q \theta_l \mathbb{E}(x_t w_{t+h-l}) \\
 &= \sum_{k=1}^p \phi_k \gamma(h-k) + \sum_{l=0}^q \theta_l \mathbb{E} \left(\sum_{k=0}^{\infty} \psi_k w_{t-k} w_{t+h-l} \right) \\
 &= \sum_{k=1}^p \phi_k \gamma(h-k) + \sum_{l=h}^q \theta_l \psi_{l-h} \sigma_w^2
 \end{aligned}$$

$$\begin{aligned}
 \phi(B)x(t) &= \theta(B)w_t \\
 x_t &= \frac{\theta(B)}{\phi(B)}w_t = \psi(B)w_t
 \end{aligned}$$

A horizontal timeline with a central point labeled t . To the left, a point is labeled $t-k$. To the right, three points are labeled $t+h-q$, $t+h-l$, and $t+h$. A double-headed arrow spans from $t+h-q$ to $t+h$, with the label $t+h-l$ centered above it.

$$\sum_{l=h}^q (\cdot) = 0, q < h$$

$$\gamma(h) - \sum_{k=1}^p \phi_k \gamma(h-k) = \sigma_w^2 \sum_{l=h}^q \theta_l \psi_{l-h}$$

Times Series Models

Autocovariance of AR(p) process

$$x_t = \sum_{k=1}^p \phi_k x_{t-k} + w_t + \sum_{l=1}^q \theta_l \cancel{x_{t-l}} \quad \text{ARMA}(p,0) \text{ process}$$

When $h < p$, these summation contains $\dots, \gamma(-1), \gamma(0), \gamma(1), \dots$

$$\gamma(h) - \sum_{k=1}^p \phi_k \gamma(h-k) = \sigma_w^2 \sum_{l=h}^0 \theta_l \psi_{l-h}$$

Consider only the case when all $\gamma(h-k)$ are unique, i.e. $h \geq p$.

$$\gamma(h) - \phi_1 \gamma(h-1) - \phi_2 \gamma(h-2) - \dots - \phi_p \gamma(h-p) = 0$$

Homogeneous difference equation.

Autocovariance of AR(p) process (2)

$$\gamma(h) - \phi_1\gamma(h-1) - \phi_2\gamma(h-2) - \dots - \phi_p\gamma(h-p) = 0$$

Plug $\gamma(h) = r^h$.

$$r^h - \phi_1r^{h-1} - \phi_2r^{h-2} - \dots - \phi_pr^{h-p} = 0$$

Since $r \neq 0$, we have

$$1 - \phi_1r^{-1} - \phi_2r^{-2} - \dots - \phi_pr^{-p} = 0$$

$$\phi(r^{-1}) = 0$$

For the causal AR process, the root of $\phi(z)$ is outside the unit circle, i.e. $|r| < 1$. Thus, $\lim_{h \rightarrow \infty} \gamma(h) = 0$.

Ex. 10: ACF of ARMA(1,2) process

Find the ACF of the following process.

$$x_t = \frac{1}{2}x_{t-1} + w_t - \frac{1}{2}w_{t-2}$$

Solution: Rewrite the process into the $\phi(B)x_t = \theta(B)w_t$.

$$(1 - 0.5B)x_t = (1 - 0.5B^2)w_t$$

$$\gamma(h) - \sum_{k=1}^1 \phi_k \gamma(h-k) = \sigma_w^2 \sum_{l=h}^2 \theta_l \psi_{l-h},$$

$$\text{where } \phi_1 = 0.5, \theta_l = \begin{cases} 1, & l = 0 \\ 0, & l = 1. \\ -0.5 & l = 2 \end{cases}$$

Ex. 10: ACF of ARMA(1,2) process

$$\gamma(h) - 0.5\gamma(h-1) = \sigma_w^2 \sum_{l=h}^2 \theta_l \psi_{l-h},$$

Find ψ_1 and ψ_2 by plugging $x_t = \psi(B)w_t$ into $\phi(B)x_t = \theta(B)w_t$

$$(1 - 0.5B)(1 + \psi_1 B + \psi_2 B^2 + \cdots)w_t = (1 - 0.5B^2)w_t$$

$$(1 + (\psi_1 - 0.5)B + (\psi_2 - 0.5\psi_1)B^2 + \cdots)w_t = (1 - 0.5B^2)w_t$$

$$\psi_1 = 0.5, \quad \psi_2 - 0.5\psi_1 = -0.5 \therefore \psi_2 = -0.25$$

Ex. 10: ACF of ARMA(1,2) process (2)

$$\gamma(h) - 0.5\gamma(h-1) = \sigma_w^2 \sum_{l=h}^2 \theta_l \psi_{l-h}, \theta_l = \begin{cases} 1, & l=0 \\ 0, & l=1 \\ -0.5 & l=2 \end{cases}, \psi_k = \begin{cases} 1, & k=0 \\ 0.5, & k=1 \\ -0.25, & k=2 \\ \dots & \end{cases}$$

$$\gamma(0) - 0.5\gamma(1) = \sigma_w^2(\theta_0\psi_0 + \theta_1\psi_1 + \theta_2\psi_2) = 1.125\sigma_w^2 \quad (\text{i})$$

$$\gamma(1) - 0.5\gamma(0) = \sigma_w^2(\theta_1\psi_0 + \theta_2\psi_1) = -0.25\sigma_w^2 \quad (\text{ii})$$

$$\gamma(2) - 0.5\gamma(1) = \sigma_w^2(\theta_2\psi_0) = -0.5\sigma_w^2 \quad (\text{iii})$$

LHS = homogeneous equation
when $h \geq \max(\mathbf{p}, \mathbf{q} + \mathbf{1})$

$$\begin{aligned} \gamma(3) - 0.5\gamma(2) &= 0 \\ \gamma(4) - 0.5\gamma(3) &= 0 \end{aligned}$$



$$\begin{aligned} h &> 2 \\ \gamma(h) - 0.5\gamma(h-1) &= 0 \end{aligned} \quad (\text{iv})$$

Ex. 10: ACF of ARMA(1,2) process (3)

$$\gamma(0) - 0.5\gamma(1) = 1.125\sigma_w^2 \quad (\text{i})$$

$$-0.5\gamma(0) + \gamma(1) = -0.25\sigma_w^2 \quad (\text{ii})$$

$$\gamma(2) - 0.5\gamma(1) = -0.5\sigma_w^2 \quad (\text{iii})$$

$$h > 2, \quad \gamma(h) - 0.5\gamma(h-1) = 0 \quad (\text{iv})$$

Solving (i)&(ii), we get $\gamma(0) = \frac{4}{3}\sigma_w^2$, $\gamma(1) = \frac{5}{12}\sigma_w^2$.

Plugging $\gamma(1)$ into (iii), we get $\gamma(2) = -\frac{7}{24}\sigma_w^2$.

Solving (iv), we get $\gamma(h) = 0.5^{h-2}\gamma(2) = -\frac{7}{6 \cdot 2^h}\sigma_w^2$.

$$\gamma(h) = \begin{cases} \frac{4}{3}\sigma_w^2, & h = 0 \\ \frac{5}{12}\sigma_w^2, & h = 1 \\ -\frac{7}{6 \cdot 2^h}\sigma_w^2, & h \geq 2 \end{cases}$$

Ex. 10: ACF of ARMA(1,2) process (4)

$$\gamma(h) = \begin{cases} \frac{4}{3}\sigma_w^2, & h = 0 \\ \frac{5}{12}\sigma_w^2, & h = 1 \\ -\frac{7}{6 \cdot 2^h}\sigma_w^2, & h \geq 2 \end{cases} \quad \Rightarrow \quad \rho(h) = \frac{\gamma(h)}{\gamma(0)} = \begin{cases} 1, & h = 0 \\ \frac{5}{16}, & h = 1 \\ -\frac{7}{2^{h+3}}, & h \geq 2 \end{cases}$$

Ex. 11: ACF of ARMA(1,1) process

$$x_t = \phi x_{t-1} + w_t + \theta w_{t-1}$$

Solution: Rewrite the process into the $\phi(B)x_t = \theta(B)w_t$.

$$(1 - \phi B)x_t = (1 - \theta B)w_t$$

$$\gamma(h) - \sum_{k=1}^1 \phi_k \gamma(h-k) = \sigma_w^2 \sum_{l=h}^1 \theta_l \psi_{l-h},$$

$$\text{where } \phi_1 = \phi, \theta_l = \begin{cases} 1, & l = 0 \\ \theta, & l = 1 \end{cases}.$$

Ex. 11: ACF of ARMA(1,1) process (1)

$$\gamma(h) - \phi\gamma(h-1) = \sigma_w^2 \sum_{l=h}^1 \theta_l \psi_{l-h}, \quad \theta_l = \begin{cases} 1, & l = 0 \\ \theta, & l = 1 \end{cases}$$

Find ψ_1 by plugging $x_t = \psi(B)w_t$ into $\phi(B)x_t = \theta(B)w_t$

$$(1 - \phi B)(1 + \psi_1 B + \psi_2 B^2 + \cdots)w_t = (1 + \theta B)w_t$$

$$(1 + (\psi_1 - \phi)B + \cdots)w_t = (1 + \theta B)w_t$$

$$\psi_1 = \phi + \theta$$

Ex. 11: ACF of ARMA(1,1) process (2)

$$\gamma(h) - \phi\gamma(h-1) = \sigma_w^2 \sum_{l=h}^1 \theta_l \psi_{l-h}, \theta_l = \begin{cases} 1, & l=0 \\ \theta, & l=1 \end{cases}, \psi_l = \begin{cases} 1, & l=0 \\ \phi + \theta, & l=1 \\ \dots & \end{cases}$$

$$\gamma(0) - \phi\gamma(1) = \sigma_w^2(\theta_0\psi_0 + \theta_1\psi_1) = (1 + \theta\phi + \theta^2)\sigma_w^2 \quad (\text{i})$$

$$\gamma(1) - \phi\gamma(0) = \sigma_w^2(\theta_1\psi_0) = \theta\sigma_w^2 \quad (\text{ii})$$

$$h > 2, \quad \gamma(h) - \phi\gamma(h-1) = 0 \quad (\text{iii})$$

$$\text{From (i)\&(ii), } \gamma(0) = \frac{1+2\theta\phi+\theta^2}{1-\phi^2} \sigma_w^2, \quad \gamma(1) = \frac{(\theta+\phi)(1+\theta\phi)}{1-\phi^2} \sigma_w^2.$$

$$\text{From (iii), } \gamma(h) = \phi^{h-1}\gamma(1), \quad h > 1.$$

Ex. 11: ACF of ARMA(1,1) process (3)

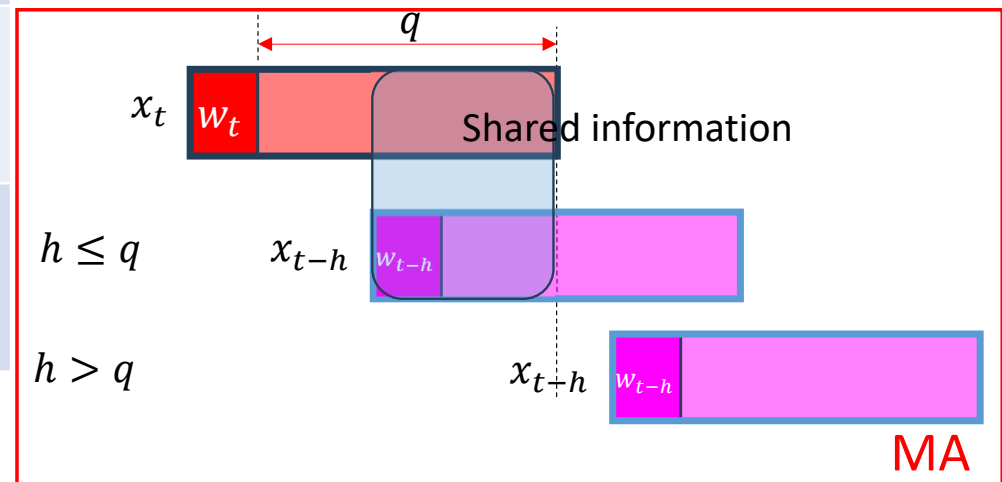
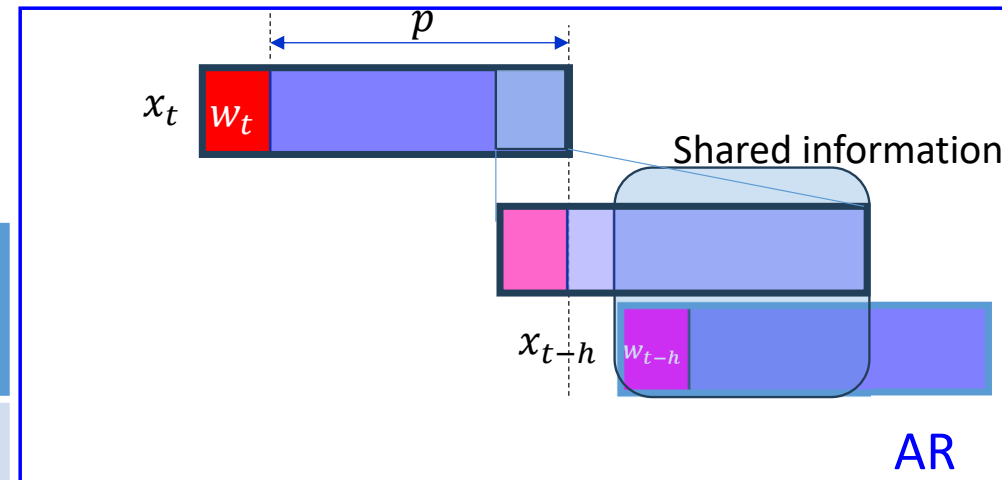
$$\gamma(h) = \begin{cases} \frac{1 + 2\theta\phi + \theta^2}{1 - \phi^2} \sigma_w^2, & h = 0 \\ \phi^{h-1} \frac{(\theta + \phi)(1 + \theta\phi)}{1 - \phi^2} \sigma_w^2, & h \geq 1 \end{cases}$$



$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \begin{cases} 1, & h = 0 \\ \phi^{h-1} \frac{(\theta + \phi)(1 + \theta\phi)}{1 + 2\theta\phi + \theta^2}, & h \geq 1 \end{cases}$$

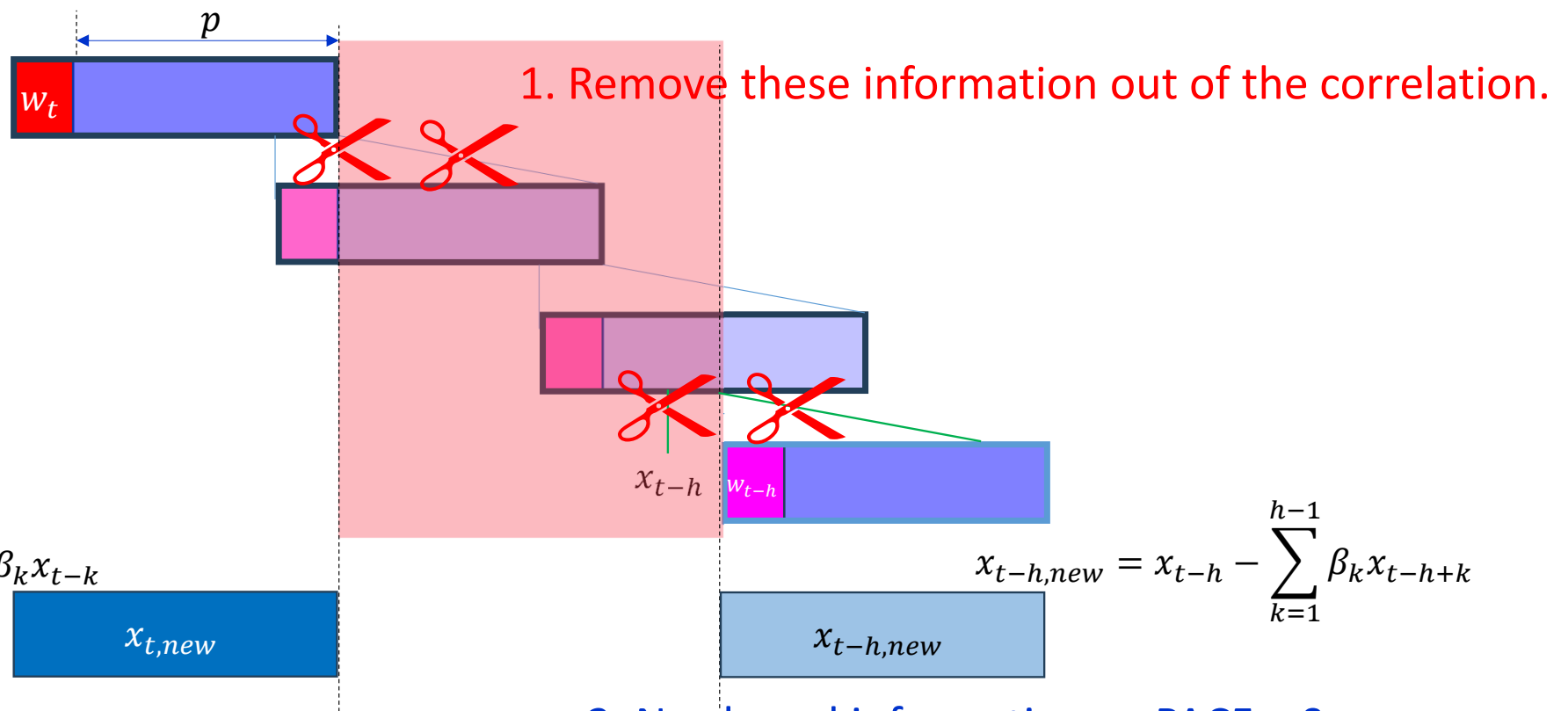
Partial summary (2)

Process	$ACF(\rho(h))$
$AR(p)$	Tails off
$MA(q)$	0 when $h > q$
$ARMA(p, q)$	Tails off



Partial autocorrelation function (PACF): Idea

Idea:



$$x_{t,new} = x_t - \sum_{k=1}^{h-1} \beta_k x_{t-k}$$

$x_{t,new}$

$$x_{t-h,new} = x_{t-h} - \sum_{k=1}^{h-1} \beta_k x_{t-h+k}$$

$x_{t-h,new}$

Information removal

Backward estimation

Linear estimation of x_t by $x_{t-1}, x_{t-2}, \dots, x_{t-h+1} = \hat{x}_t$

$$x_{t,new} = x_t - \sum_{k=1}^{h-1} \beta_k x_{t-k}$$

Forward estimation

Linear estimation of x_t by $x_{t+1}, x_{t+2}, \dots, x_{t+h-1} = \hat{x}_t$

$$x_{t,new} = x_t - \sum_{k=1}^{h-1} \beta_k x_{t+k}$$

PACF: Definition

“The partial autocorrelation function (PACF) of a stationary process x_t , denoted by ϕ_{hh} , for $h = 1, 2, \dots$ is

$$\phi_{11} = \text{corr}(x_{t+1}, x_t) = \rho(1)$$

and

$$\phi_{hh} = \text{corr}(x_{t+h} - \hat{x}_{t+h}, x_t - \hat{x}_t), h \geq 2$$

Ex. 12: PACF of AR(1)

$$x_t = w_t + \phi x_{t-1}$$

Solution:

$$\phi_{11} = \text{corr}(x_{t+1}, x_t) = \rho(1)$$

From Ex.1, we have $\rho(h) = \phi^h$ so $\phi_{11} = \phi$

$$\phi_{hh} = \text{corr}(x_{t+h} - \hat{x}_{t+h}, x_t - \hat{x}_t), h \geq 2$$

Case $h = 2$, $\hat{x}_{t+2} = \beta_1 x_{t+1}$ and $\hat{x}_t = \beta'_1 x_{t+1}$. Find β_1, β'_1 that give minimum mean square error (MSE).

Ex. 12: PACF of AR(1) (2)

$$\begin{aligned}\beta_1 &= \arg \min_{\beta_1} E(x_{t+2} - \hat{x}_{t+2})^2 = \arg \min_{\beta_1} E(x_{t+2} - \beta_1 x_{t+1})^2 \\ E(x_{t+2} - \beta_1 x_{t+1})^2 &= E(x_{t+2}^2 - 2\beta_1 x_{t+2} x_{t+1} + \beta_1^2 x_{t+1}^2) \\ &= \gamma(0) - 2\beta_1 \gamma(1) + \beta_1^2 \gamma(0)\end{aligned}$$

Take the derivative with β_1 and set the derivative to 0.

$$\begin{aligned}\frac{d}{d\beta_1} [E(x_{t+2} - \beta_1 x_{t+1})^2] &= \frac{d}{d\beta_1} [\gamma(0) - 2\beta_1 \gamma(1) + \beta_1^2 \gamma(0)] \\ 0 &= [0 - 2\gamma(1) + 2\beta_1 \gamma(0)] \\ \beta_1 &= \frac{\gamma(1)}{\gamma(0)} = \rho(1) = \phi\end{aligned}$$

Ex. 12: PACF of AR(1) (3)

$$\beta'_1 = \arg \min_{\beta'_1} E(x_t - \hat{x}_t)^2 = \arg \min_{\beta'_1} E(x_t - \beta'_1 x_{t+1})^2$$

$$\begin{aligned} E(x_t - \beta'_1 x_{t+1})^2 &= E(x_t^2 - 2\beta'_1 x_t x_{t+1} + \beta_1'^2 x_{t+1}^2) \\ &= \gamma(0) - 2\beta'_1 \gamma(1) + \beta_1'^2 \gamma(0) \end{aligned}$$

This is the same error function that we use to find β_1 , so

$$\beta'_1 = \beta_1 = \phi.$$

$$\phi_{22} = \text{corr}(x_{t+2} - \phi x_{t+1}, x_t - \phi x_{t+1}) = \text{corr}(w_{t+2}, x_t - \phi x_{t+1}) = 0$$

$$x_t = w_t + \phi x_{t-1}$$

Ex. 12: PACF of AR(1) (4)

Case $h = 3$, $\hat{x}_{t+3} = \beta_1 x_{t+2} + \beta_2 x_{t+1}$ and $\hat{x}_t = \beta'_1 x_{t+1} + \beta'_2 x_{t+2}$.

Find β_k, β'_k that give minimum mean square error (MSE).

$$\begin{aligned} E[x_{t+3} - \hat{x}_{t+3}]^2 &= E[x_{t+3} - (\beta_1 x_{t+2} + \beta_2 x_{t+1})]^2 \\ &= E[x_{t+3}^2 - 2(\beta_1 x_{t+2} + \beta_2 x_{t+1})x_{t+3} + (\beta_1 x_{t+2} + \beta_2 x_{t+1})^2] \\ &= \gamma(0) - 2(\beta_1 \gamma(1) + \beta_2 \gamma(2)) + \\ &\quad E[(\beta_1^2 x_{t+2}^2 + 2\beta_1 \beta_2 x_{t+2} x_{t+1} + \beta_2^2 x_{t+1}^2)] \\ &= (1 + \beta_1^2 + \beta_2^2) \gamma(0) - 2\beta_1 (1 - \beta_2) \gamma(1) - 2\beta_2 \gamma(2) \end{aligned}$$

Ex. 12: PACF of AR(1) (5)

$$E[x_{t+3} - \hat{x}_{t+3}]^2 = (1 + \beta_1^2 + \beta_2^2)\gamma(0) - 2\beta_1(1 - \beta_2)\gamma(1) - 2\beta_2\gamma(2)$$

Take the partial derivative with β_1, β_2 and set the derivative to 0.

$$0 = 2\beta_1\gamma(0) - 2(1 - \beta_2)\gamma(1) \quad \Rightarrow \quad \rho(1) = \beta_1 + \rho(1)\beta_2$$

$$0 = 2\beta_2\gamma(0) + 2\beta_1\gamma(1) - 2\gamma(2) \quad \Rightarrow \quad \rho(2) = \rho(1)\beta_1 + \beta_2$$

$$\text{So } \beta_1 = \frac{\rho(1)(1-\rho(2))}{1-\rho(1)^2} = \phi, \beta_2 = \frac{\rho(2)-\rho(1)^2}{1-\rho(1)^2} = 0$$

Ex. 12: PACF of AR(1) (6)

$$\begin{aligned} E[x_t - \hat{x}_t]^2 &= E[x_t - (\beta'_1 x_{t+1} + \beta'_2 x_{t+2})]^2 \\ &= E[x_t^2 - 2(\beta'_1 x_{t+1} + \beta'_2 x_{t+2})x_t + (\beta'_1 x_{t+1} + \beta'_2 x_{t+2})^2] \\ &= \gamma(0) - 2(\beta'_1 \gamma(1) + \beta'_2 \gamma(2)) + \\ &\quad E[(\beta_1'^2 x_{t+1}^2 + 2\beta'_1 \beta'_2 x_{t+1} x_{t+2} + \beta_2'^2 x_{t+2}^2)] \\ &= (1 + \beta_1'^2 + \beta_2'^2) \gamma(0) - 2\beta'_1 (1 - \beta'_2) \gamma(1) - 2\beta'_2 \gamma(2) \end{aligned}$$

This is the same error function that we use to find β_1 and β_2 , so

$$\beta'_1 = \beta_1 = \phi, \quad \beta'_2 = \beta_2 = 0.$$

Ex. 12: PACF of AR(1) (7)

$$\phi_{33} = \text{corr}(x_{t+3} - \phi x_{t+2}, x_t - \phi x_{t+1}) = \text{corr}(w_{t+3}, x_t - \phi x_{t+1}) = 0$$

The diagram illustrates the substitution of the AR(1) model equation into the PACF formula. A green box highlights the term $x_{t+3} - \phi x_{t+2}$ in the first part of the equation. A blue line connects this box to a green box containing the equation $x_t = w_t + \phi x_{t-1}$. Another blue line connects this central box to a green box highlighting the term w_{t+3} in the second part of the equation.

We can continue and get $\phi_{hh} = 0$ when $h > 1$.

PACF of MA

For $AR(p)$, we have $\phi_{hh} = 0$ when $h > p$. However, $MA(q)$ is the infinite AR process. **So PACF of MA is not zero but converges to zero** at large h .

Summary

Process	ACF($\rho(h)$)	PACF(ϕ_{hh})
AR(p)	Tails off	0 when $h > p$
MA(q)	0 when $h > q$	Tails off
ARMA(p, q)	Tails off	Tails off